

1801
Joseph Jacquard's loom uses a program (run by punched cards) to weave fabric.



Vacuum tube

1906
The vacuum tube, an essential part of modern computers, is invented.

1943
British engineer Thomas Flowers builds Colossus: the first electronic, digital computer.

1946
ENIAC is created – the world's first general-purpose electronic computer. It weighs 100 tonnes and contains 18,000 electronic switches.

1962
Computer company IBM sets up SABRE, a system that connects up 1,500 computer terminals.



Microchip

1971
Intel 404, the first single-chip microprocessor, is invented.

1995
A USB is used for the first time to connect other devices to a computer.

USB

1822
Charles Babbage's engine has an input, a memory, and a number-cruncher (processor).



Babbage Engine

1886
Herman Hollerith builds the first punched-card tabulating and sorting machine.



Herman Hollerith

1941
German Konrad Zuse designs the Z3, the world's first working, programmable, fully automatic digital computer.

1947
The transistor is invented. It would allow electronic devices to become much smaller.



Transistor

1976
The world's first supercomputer, CRAY-1, is built.

1981
IBM launches a PC that uses MS-DOS as an operating system.

1991
The World Wide Web is made publicly available.

2014
The first 8-terabyte hard drive is released.

2015

THE INTERNET

The Internet is a computer network that stretches around the world, linking most computers on the planet. Every computer has its own Internet or IP address, so that digital things (such as email) can be sent to or from it.

3. Separate packets travel across different routes over the Internet

2. Each packet is labelled with the IP destination address

1. Sender's computer breaks photo into many tiny digital pieces, or "packets"

4. Pieces are reassembled at the end

5. Receiver sees the final picture exactly as sent

SENDING A PHOTO

When you send something like a photo by email, the Internet breaks it into small pieces then reassembles it.

WHAT WE DO ONLINE?

We now use the Internet for all sorts of activities where we want to connect with someone else – either for fun or for business.



EMAIL

Emails are an instant way to send a digital letter.



GAMES

We can play games with distant friends via the Internet.



SHOPPING

We can buy things online from anywhere in the world.



SOCIAL NETWORKING

Groups of people can communicate easily online.

COMPUTERS EVERYWHERE

Computers are used in all sorts of devices, from personal music players and phones to microwave ovens and surveillance cameras.



PORTABLE GPS



PORTABLE MEDIA PLAYER



SMARTPHONE



DIGITAL TELESCOPE



DIGITAL RADIO



SURVEILLANCE CAMERA



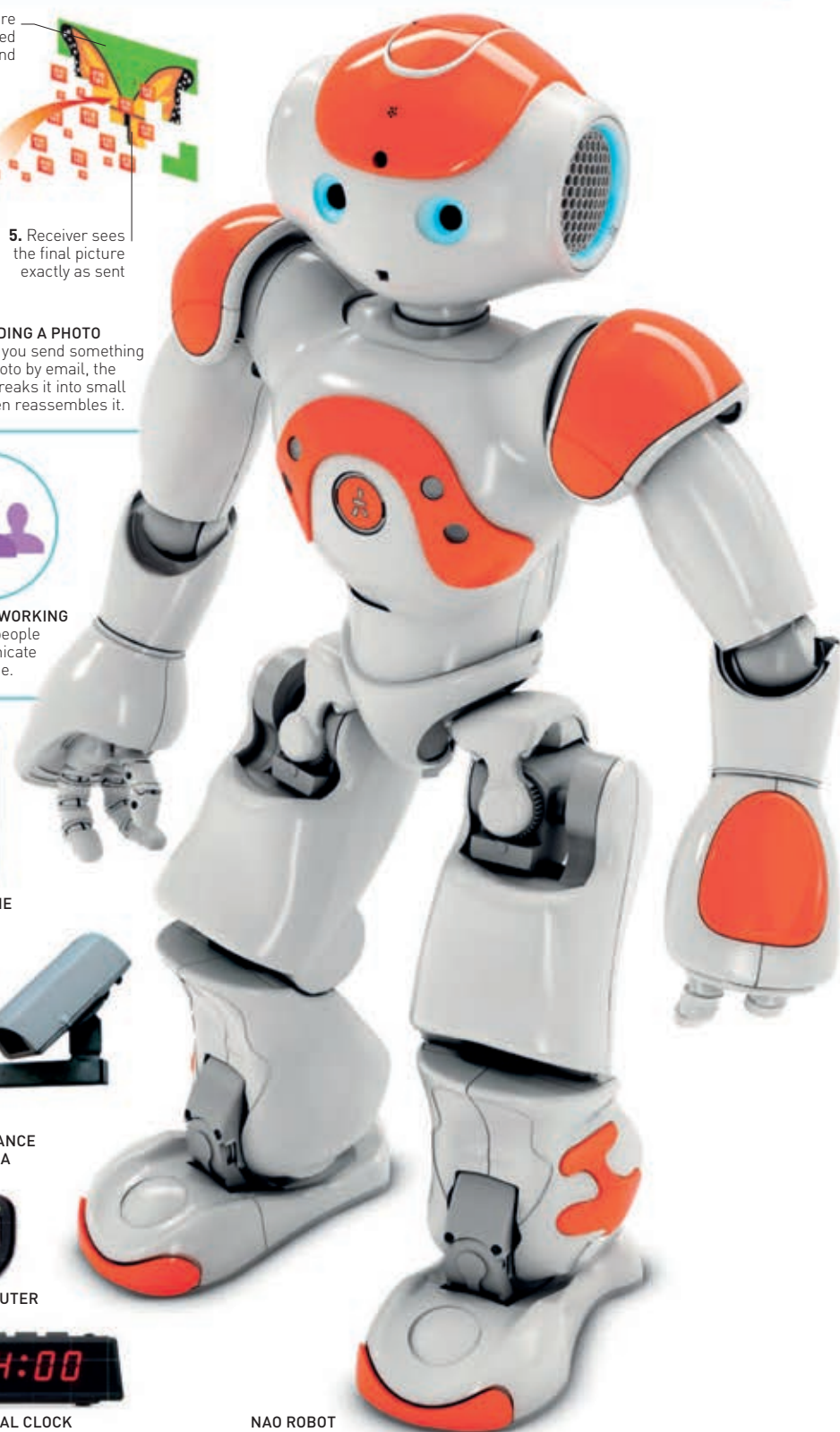
MICROWAVE OVEN



CYCLE COMPUTER



DIGITAL CLOCK



NAO ROBOT